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CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 5 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

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Introduction

To measure physical phenomena in the real world with a computer, we need to use the electrical properties of a circuit. This often includes resistance, inductance, and capacitance. In this lab, we will use capacitance to measure the volume of water in a container and inductance to identify different types of metal to gain experience with measurement.

Methods

To measure Capacitance in a real-world setting, we are using it to measure water level. This provides a practical method for measuring water volume, especially in cases where direct measurement is not feasible or would be intrusive. The approach relies on the principle that water has a distinct dielectric constant. By tracking changes in capacitance as the sensor interacts with water, we can determine the water volume along a copper strip.

For a real-world example of inductance, we are tracking different metal types and metallic objects. To do this we are using a coil to observe how different metals alter the magnetic field around the coil. This response depends on the metal's magnetic permeability and electrical conductivity.

Initial Experiments

We used one of the plastic containers provided to attach electrodes to its outside, forming a capacitor for liquid level measurement. Then, we built an integrator circuit by connecting a resistor between a 3.3V source and the capacitor (Vcap node), with the other end of the capacitor grounded. We added a transistor to allow software-controlled de-integration of the capacitor's charge, as demonstrated in class. We then connected Vcap to the input of the controller's comparator 0 (C0-) and configured an internal reference to drive C0+ (Vref).

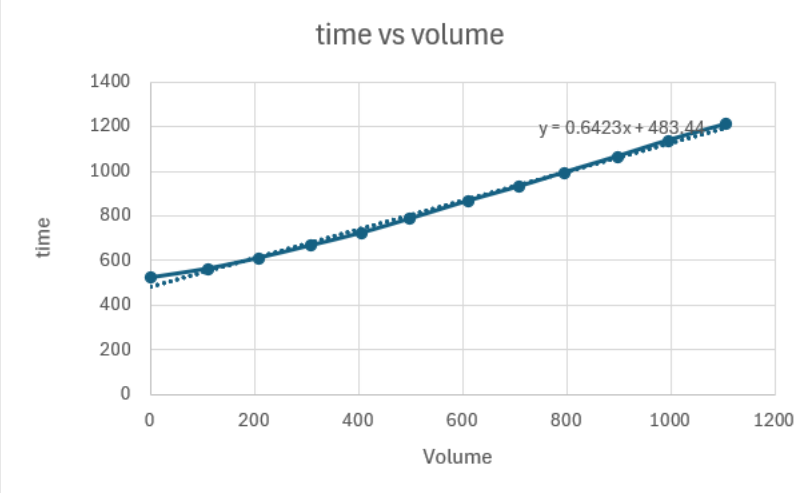
We then wrote code to briefly activate the transistor to de-integrate the capacitor's charge, measuring the time it took for the comparator to change state (when Vcap > Vref). Using empirical data, we developed a method to convert this time measurement into milliliters of water in the container.

PUTTY Captures:

```
Time:        546 (ticks)mL:          0 (mL)
Time:        602 (ticks)mL:          0 (mL)
Time:        666 (ticks)mL:        284 (mL)
Time:        690 (ticks)mL:        322 (mL)
Time:        722 (ticks)mL:        372 (mL)
Time:        762 (ticks)mL:        434 (mL)
Time:        810 (ticks)mL:        509 (mL)
Time:        866 (ticks)mL:        596 (mL)
Time:        914 (ticks)mL:        671 (mL)
Time:        914 (ticks)mL:        671 (mL)
Time:        922 (ticks)mL:        683 (mL)
Time:        922 (ticks)mL:        683 (mL)
Time:        922 (ticks)mL:        683 (mL)
Time:        922 (ticks)mL:        683 (mL)
Time:        922 (ticks)mL:        683 (mL)
Time:        914 (ticks)mL:        671 (mL)
```

This is at a constant pour from 0mL to 680mL with a decent level of accuracy.

Using Excel and plotting the values we measured we obtained this graph:



Afterwards we built a metal detector circuit using a large loop of wire and a Colpitts oscillator connected to the WT1CCP0 input. Placing metal over the loop and noting the change in frequency resulting from the eddy currents in the metal and the resulting change in inductance.

Lastly, we wrote code to determine the average frequency of the loop. When the metal is brought near to the coil (like a car riding over the inductive loop at a traffic intersection), pulsing an LED on briefly showing that metal is detected.

Using the unadjusted value of the wide timer we found the following ranges for different metallic surfaces:

Scale: 95000-97000

Power supply: 98000-100000

Oscilloscope: 125000-1290000

```
Frequency:  110619 (Hz)
Frequency:  127931 (Hz)
Frequency:  127974 (Hz)
Frequency:  127920 (Hz)
Frequency:  127928 (Hz)
Frequency:  128050 (Hz)
Frequency:  128121 (Hz)
Frequency:  128173 (Hz)
Frequency:  128007 (Hz)
Frequency:   98506 (Hz)
Frequency:   93627 (Hz)
Frequency:   96510 (Hz)
Frequency:   98142 (Hz)
Frequency:   97477 (Hz)
Frequency:   93553 (Hz)
Frequency:   93558 (Hz)
Frequency:   93559 (Hz)
Frequency:   93554 (Hz)
Frequency:   93559 (Hz)
Frequency:   93598 (Hz)
Frequency:   96862 (Hz)
Frequency:   97946 (Hz)
Frequency:   97959 (Hz)
Frequency:   97968 (Hz)
Frequency:   97976 (Hz)
Frequency:   97357 (Hz)
Frequency:   99202 (Hz)
Frequency:   99215 (Hz)
Frequency:   99225 (Hz)
Frequency:   99236 (Hz)
Frequency:   99242 (Hz)
```

This is a terminal output of the measured frequency as the loop passed different objects.

Conclusions

Tn this lab we developed a non-contact level sensing approach to water tanks that can be applied to several types of liquid storage. Additionally, by comparing the measured inductance change against known values for different metals, we can identify the type of metal. As a result of knowledge gained in this lab, we now better understand capacitance and inductance. This has given us a better idea of how to create sensors for many different applications.

Circuit Built:

